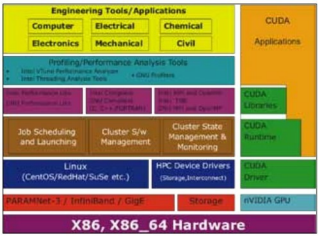

Figure 1: Hierarchical diagram

interaction with the HPC hardware. Coupled with three revolutionary HPC products from C-DAC, Onama hopes to change the perspective of today's students, academicians, researchers, and engineers. To help engineering students understand the working of HPC and its benefits, Onama is being introduced in various Indian engineering colleges.

A C-DAC initiative

The Centre for Development of Advanced Computing, popularly known as C-DAC, is an autonomous body primarily involved in the design, development and deployment of advanced IT-based solutions in India. In fact, C-DAC is the pioneer in the field of supercomputers in India, and has served numerous central and state government organisations. Its activities are funded by the Ministry of Communications and Information Technology, Government of India. C-DAC is dedicated to making HPC a tool to increase the competitiveness of Indian organisations and manpower, worldwide. To achieve this, C-DAC is focusing on strengthening HPC usage at the grass root level. Today's innovative engineering students are the ones who will be responsible for tomorrow's technological developments. To enable them to fuel the HPC momentum and make their future endeavours more worthwhile, C-DAC has started an initiative to help educational institutes create HPC facilities within their premises. These facilities will act as a nucleus for all HPC-related activities in the institute, and will be scaled up to meet future needs. For educational institutes, C-DAC's product line-up consists of three HPC cluster options. Nisha Kurkure, team leader, HPCS (HPC Solutions Group), C-DAC, says, "Onama is an integrated package that opens a new door to future technocrats, providing them a quantum leap in developing a firm understanding through High Performance Computing in several engineering disciplines."

This project includes the installation and commissioning of the HPC facility. Training on parallel processing and usage of the HPC facility will also be provided, to kick-start productive use of the infrastructure at the earliest. This offer is being provided by the HPCS, which has its base at the C-DAC headquarters situated in Pune. HPCS has 20 years of experience in HPC design and deployment. The project is being overseen by a team of application developers dedicated to the development and porting of applications on parallel platforms. The HPCS technical support team keeps abreast with the latest emerging technologies in the field. These technologies will eventually be transferred to educational institutes through Onama, thus exposing students to the most current advancements in real time

Option 1:

DL 1000- based HPC cluster
Master node type: DL 170h
Switch: 24 port GigE
Compute node type: DL 170h
Number of compute nodes: 3

Option 2:

Blades-based HPC cluster
Master node type: BL 280cG6
Disk space: 2 × 500 GB disks
Switch: Blade GigE
Compute node type: BL 280cG6
Number of compute nodes: 3

Option 3:

DL 160-based HPC cluster
Master node type: DL 180cG6
Disk space: 2 × 250 + 4 × 500 GB disks
Switch: 24 port GigE
Compute node type: DL 160cG6
Number of compute nodes: 7

Applications for different engineering disciplines

Onama is an integrated package designed to help aspiring technocrats develop a firm understanding of HPC in the various engineering disciplines. It consists of hand-picked tools for all major engineering disciplines. Figure 2 shows the official Onama installer, which allows users from different branches of engineering to select from the applications included. These applications are viable alternatives to traditional industry-standard software, and are actually used by a number of companies as well as individuals. Sandeep Agrawal, team leader, HPCS Group, C-DAC, says, "Onama provides engineers with the tools that have close affinity with the practical simulation, hence making the learning process a lot more applied. Many of the tools are GUI-based and hence reduce the learning curve."